

PRATEEK KUMAR

SOFTWARE ENGINEER | BACKEND & AI SYSTEMS

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Backend engineer with 2+ years building production systems end to end. Real-time services on Node.js and WebSockets, multi-tenant RAG infrastructure on FastAPI and pgvector, and native Android device management at Device Owner level. I architected and shipped the company's device fleet control platform end to end, from first design through production across 7 deployment targets. Consumer work I have shipped has drawn 14.5M views and 216K shares. GATE 2024 qualified.

PROFESSIONAL EXPERIENCE

Phygitech | Software Engineer, Backend and Systems

03/2025 – Present

- Architected and shipped the company's real-time device fleet control platform on my own, from first design through to production. Next.js 16 on a custom Node.js WebSocket server, PostgreSQL with Drizzle, and a bundled MediaMTX streaming server, deployed on AWS EC2 and packaged for **7 targets** across cloud, desktop and Android.
- Designed the real-time control protocol that drives the whole fleet. The platform is the server, the on-headset MDM app the client, over WebSocket on LAN or cloud. Exactly one command channel stays live at a time, so a device can never receive conflicting commands from two sources, and operators switch a fleet between LAN and cloud on the fly.
- Runs a full class of **10 headsets** in a school, every one streaming live at roughly 0.5s WebRTC latency over LAN, and stress tested to **50+ concurrent streaming connections** served from mid-tier hardware. Screencasting picks one of three transports automatically, with HLS as fallback.
- Built the security model so the client is never trusted. Role-based permission tiers, cryptographically signed offline tokens for disconnected use, and a CI guard that fails the build when access-control rules break.
- Built its on-device counterpart, an enterprise Android MDM in Kotlin running as Device Owner through the OEM SDK on our own headsets, where silent install and uninstall, shutdown and reboot all work. One APK also covers stock consumer hardware like Oculus Quest, which will not grant Device Owner, by detecting what the platform allows and degrading cleanly.
- Built the in-house content encryption and decryption layer. Decryption runs inside the Android playback path with no measurable frame impact, which let us drop third-party DRM licensing across the whole content library.
- Set up how the team uses AI for development. Started on GitHub Copilot, then moved everyone onto agentic tools like Claude Code, Codex and Antigravity, with custom MCP server integrations wired into their editors, and trained colleagues on all of it. Measured roughly **40% time saved** across the team.
- Shipped **100+ Unity learning modules** for Grades 1 to 12 (CBSE/NCERT), built for three targets: our OEM headsets through their native SDK, OpenXR headsets, and WebGL for the browser. Wrote native Hindi, Gujarati, Marathi and Odia text renderers alongside them, and refactored the OEM SDK so Unity Editor tooling runs on macOS.

Wrexa | Software Developer, Interactive Applications

05/2024 – 03/2025

- Shipped 11 interactive projects for brands including Samsung and Cadbury, working across Unity, PlayCanvas and Lens Studio in C#, C++ and JavaScript.
- Built a multi-user 3D metaverse for Samsung in PlayCanvas and WebGL, with custom avatars, live multi-user interaction, and performance tuning that held up across desktop and mobile browsers.
- Built the Oreo Space Dunk AR lens for Cadbury in Snap AR, with face and hand tracking and physics-based gameplay, shipped to Snapchat users worldwide.
- Led a team of developers and 3D artists. Ran requirement gathering directly with clients, turned briefs into technical scope, and kept delivery on schedule.
- Owned the browser performance work on the Samsung metaverse, which is what let it hold frame rate on mid-range Android handsets where the first build could not.

DRDO-INMAS | Software Developer (R&D Intern)

05/2023 – 07/2023

- Built a Unity VR training simulation for a defence research lab, instrumented with a real-time data analysis layer that captured trainee behaviour mid-run rather than only at the end. It cut completion time 20% against the method already in use.
- Wrote the telemetry pipeline behind it, so the lab could compare performance across sessions instead of scoring one final result.

EDUCATION

B.Tech, Computer Science & Engineering

2020 – 2024

Ch. Ranbir Singh State Institute of Engineering & Technology

- GATE 2024 qualified (top 14% of 124,000 candidates)

Class XII

2020

R. C. Memorial Sr. Sec. School, Jhajjar, Haryana

Class X

2018

Saarthi Public Sr. Sec. School, Jhajjar, Haryana

PROJECTS

Cortexa, Multi-Tenant AI Knowledge Platform

FastAPI • PostgreSQL + pgvector • Redis • Celery • Next.js 16 • Docker

Businesses upload their own domain content and the assistant answers strictly from that tenant's data, so nothing leaks across tenants. FastAPI backend, pgvector for embedding search, Redis and Celery for ingestion, JWT RS256 auth, and swappable LLM providers across Groq, Claude, OpenAI and Ollama.

[Hue & Seek](#) ↗

Lens Studio • TypeScript • Snap AR • WebAR • bit.ly/hue-and-seek

Rebuilds the core loop of the hit mobile game Meccha Chameleon as a hosted WebAR lens: paint a 3D character to blend into the background before the timer runs out. It fits under 8 MB and runs straight in the browser with no install at all, where the original is a full app download. Samples real background pixels through procedural textures, paints by raycasting dabs onto capsule colliders, and scores with my own blend, creativity and coverage algorithms.

[Snap AR Lens Portfolio](#) ↗

Lens Studio • JavaScript • Snap AR • bit.ly/prateek-snap-portfolio

37 shipped lenses drawing **14.5M views** and **216K shares**. Brand campaigns for Castrol, Colgate and MinuteMaid, festival activations for Diwali, Navratri, Dussehra and Raksha Bandhan, and gameplay lenses built on face tracking, multi-object detection, VFX and physics. Top performers: SpaceShooterAR and Highway Chase.

SKILLS

Languages

TypeScript/JavaScript • Kotlin • C# • Python • C • C++ • SQL • Swift

AI-Assisted Engineering

GitHub Copilot • Claude Code • Codex • Antigravity • MCP (Model Context Protocol) • Custom MCP Servers • Agentic Coding Workflows • Prompt Engineering

Backend & Full Stack

Node.js • FastAPI • Next.js (App Router) • React • PostgreSQL • Redis • WebSocket protocol design • REST API design • TCP binary protocols • pgvector • Drizzle ORM • Celery • WebRTC • MediaMTX • Tauri

Mobile & Systems

Android (Kotlin) • Device Owner / MDM • DevicePolicyManager • AIDL • PackageInstaller • Room • Foreground Services

Cloud & DevOps

AWS (EC2, Elastic IP, Security Groups) • Docker • Linux • CI/CD Pipelines • Multi-platform Build & Packaging • Git/GitHub

XR & Graphics

Unity (AR/VR/2D/3D) • OpenXR • WebGL • PlayCanvas • Lens Studio • Blender

Concepts

Backend Engineering • System Design & Architecture • Real-Time & Distributed Systems • RAG & Vector Search • RBAC / Access Control • Applied Cryptography • Multi-Tenant Architecture • LLM Integration • Data Structures & Algorithms (DSA) • OOP • Requirement Gathering